

AN INTELLIGENT SYSTEM FOR DETECTION, TRANSLATION AND HISTORICAL MAPPING OF ANCIENT INSCRIPTIONS



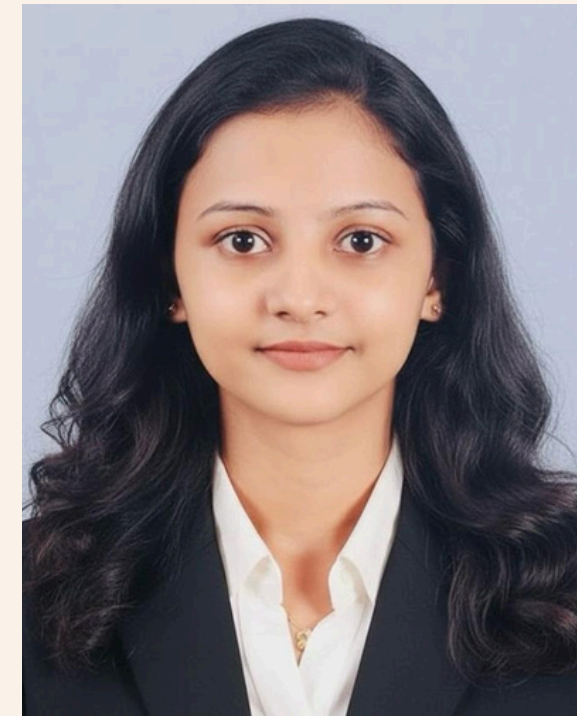
Meet our team



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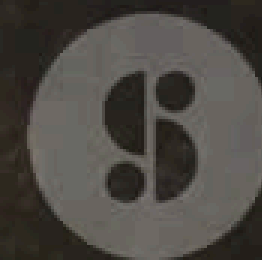
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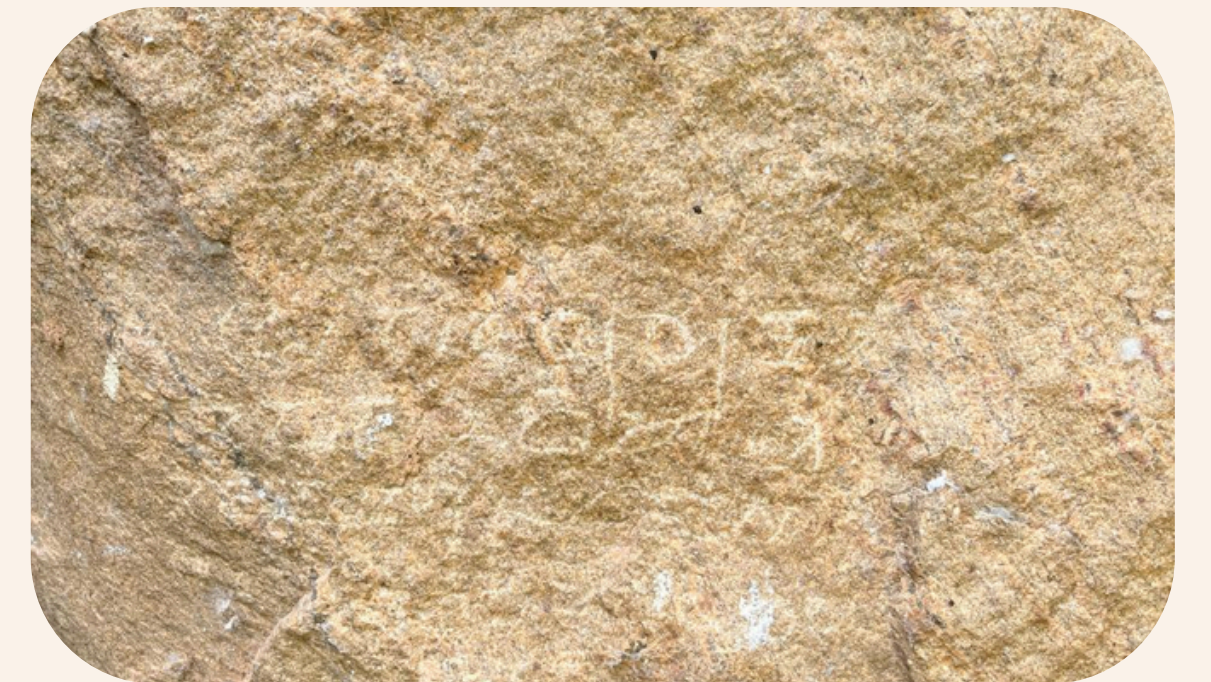
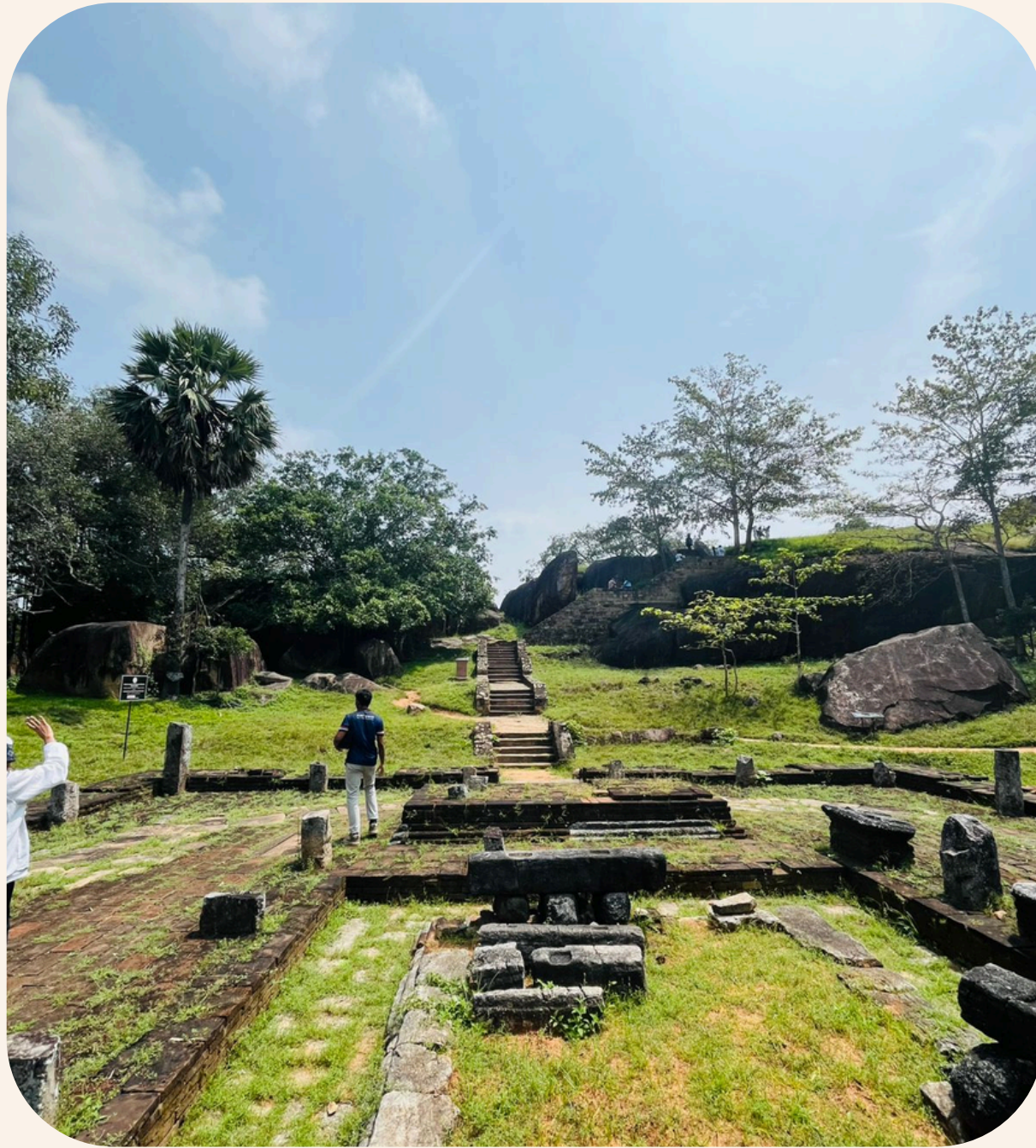
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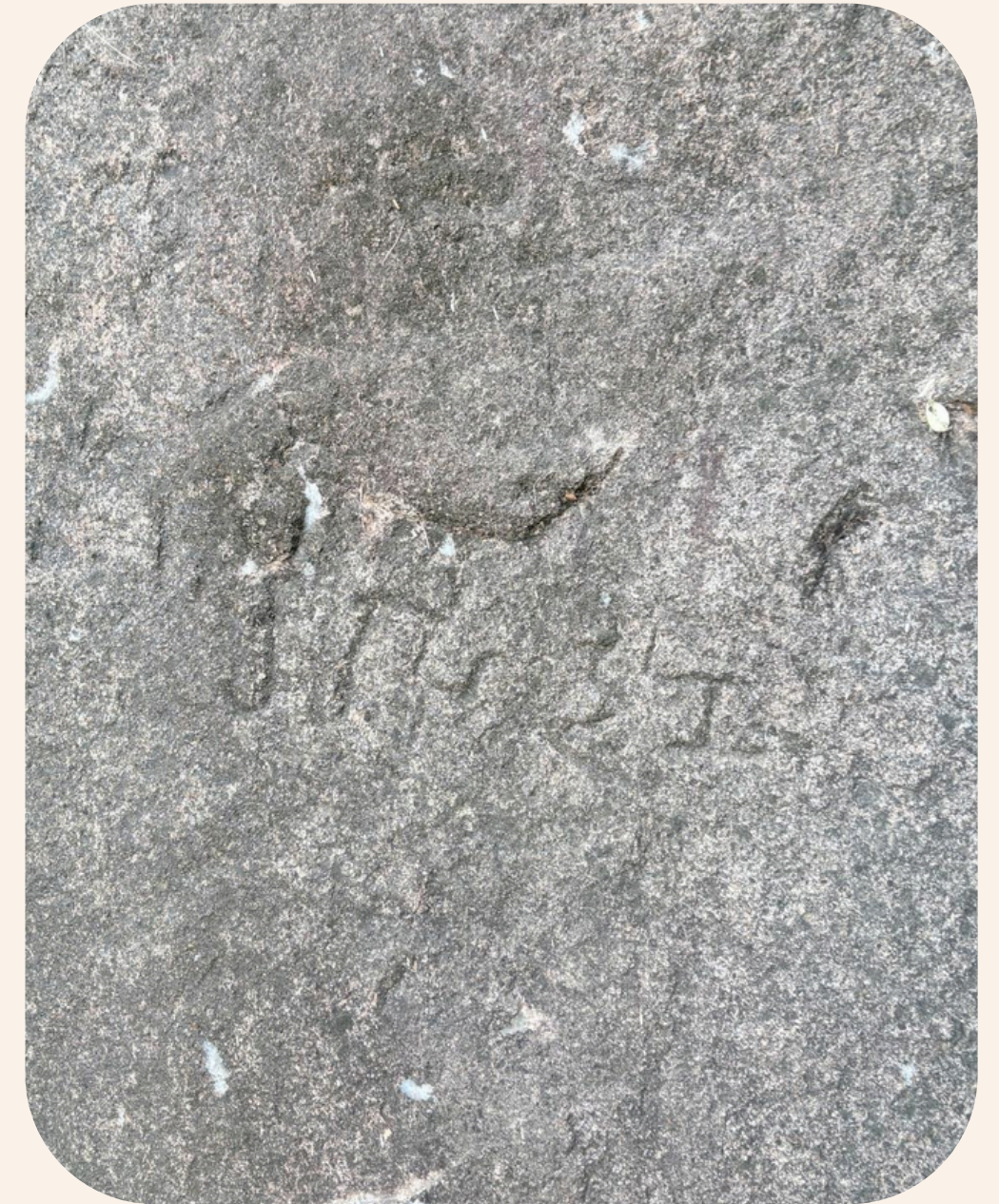
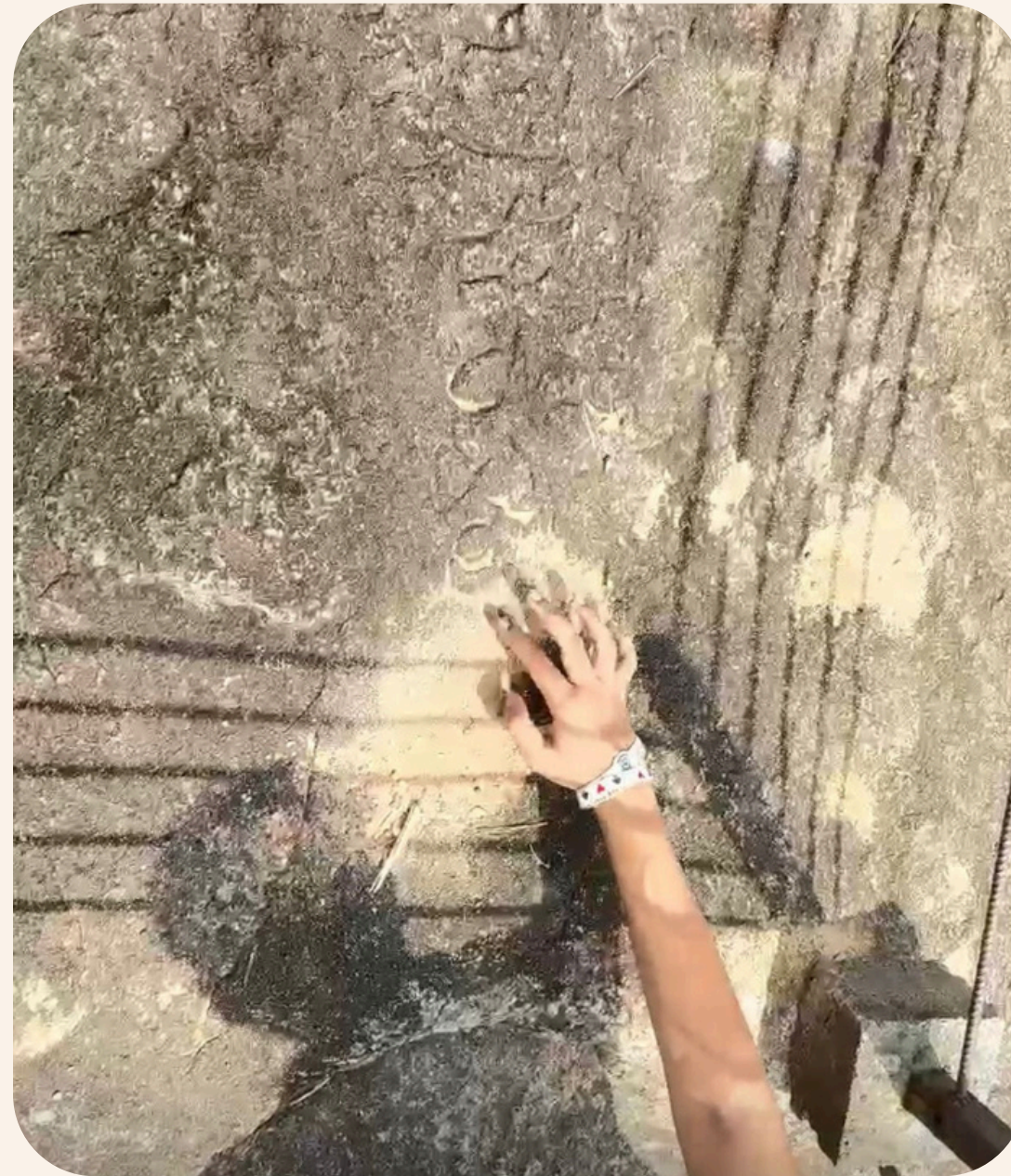
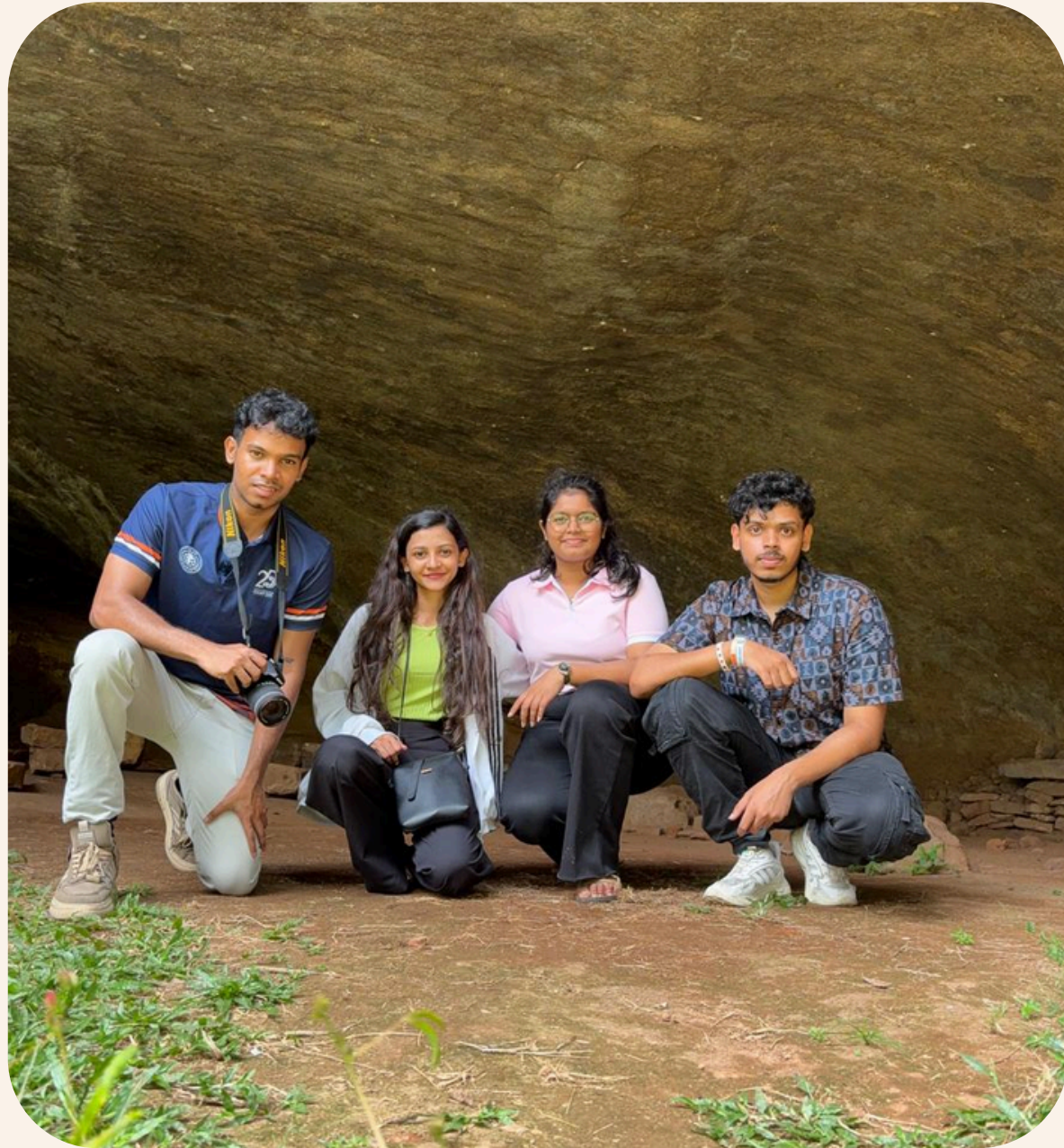


Storyblocks

Field Visit to Anuradhapura – Inscription

Data Collection





Problem & Stakeholders

Context

- Ancient inscriptions are an important source of Sri Lanka's historical information. However, many inscriptions are damaged and difficult to interpret.

Stakeholders

Primary Users

- Archaeologists
- Historians
- Epigraphists

Secondary Users

- Researchers
- Students
- General Public

Observed Issues

- Ancient inscriptions are eroded, faded, and hard to read.
- Manual transcription and translation require significant expert effort.
- Existing OCR systems mainly support modern scripts, not Early Sinhala.
- Limited datasets and tools for automated inscription analysis.

Impact

- Loss of valuable historical knowledge due to unreadable inscriptions.
- Slow research processes caused by manual transcription and interpretation.
- Limited access to inscription data for researchers and the public.
- Challenges in preserving Sri Lanka's cultural heritage digitally.

Research Gap

Existing systems have several limitations:

- OCR systems mainly focus on modern scripts not ancient inscriptions.
- Limited datasets are available for training models on ancient Sinhala characters.
- Lack of complete pipeline from detection → restoration → recognition → Segmentation → translation

Main Objective

- To develop an AI-based system that detects, restores, recognizes, segments, and translates ancient inscriptions, enabling researchers to efficiently digitize, interpret, and analyze historical texts.

Solution Overview

Solution Summary

Silent Script is an AI-powered system designed to detect, recognize, restore translate, and analyze ancient inscriptions from Sri Lanka.

Key Features Delivered

- FR1 : Image Sharpening & Character Segmentation
- FR2 : Character Recognition
- FR3 : Word Segmentation & Translation
- FR4 : Multimodel Restoration & Contextual conception

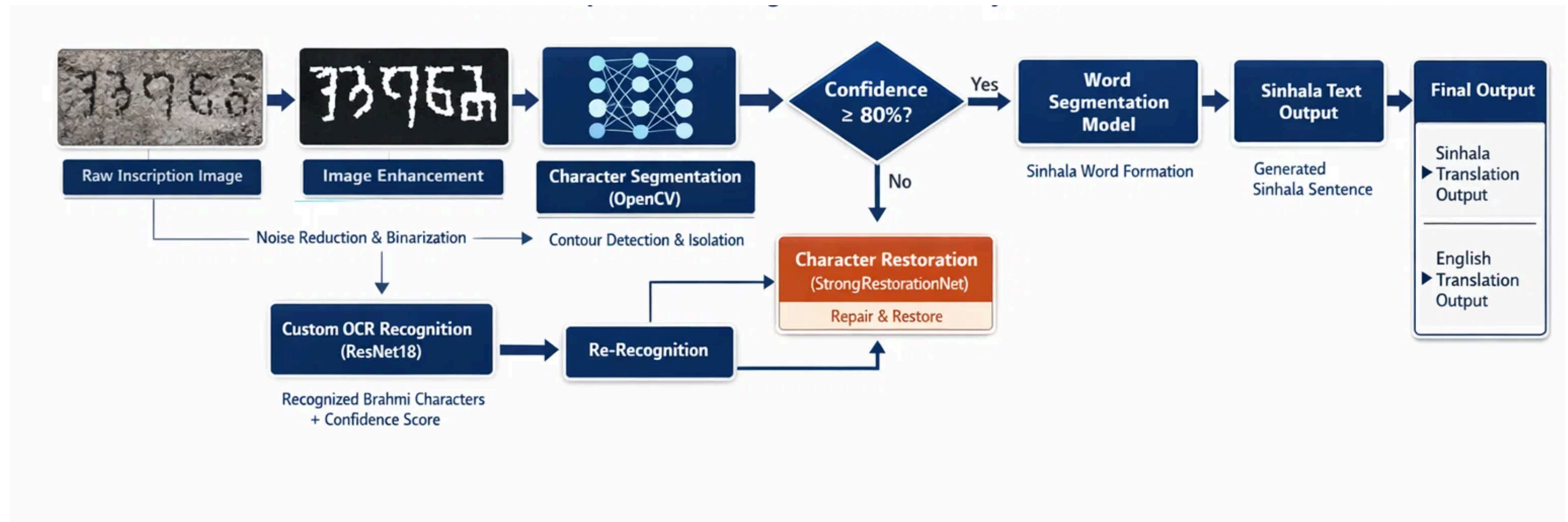
Non Functional Focus

- High accuracy in recognition
- Efficient processing performance
- User-friendly interface
- Reliable data handling

**Current
Completion Status**

~75% + Complete ⁹

System Architecture & Integration



Commercialization

Target Users / Market

- Historians, archaeologists, and epigraphists
- Archaeological Department of Sri Lanka
- Universities and research institutions
- Students and Tourists

Value Proposition

- Helps detect, recognize, and translate ancient inscriptions automatically
- Saves time and effort compared to manual analysis
- Makes inscription data easier to access and study

What Makes us Unique

- Specialized AI models for ancient Sinhala/Brahmi inscriptions
 - End-to-end pipeline from image → recognition → translation
- Helps digitize and preserve historical inscriptions

Cost Recovery / Revenue Strategy

- Subscription-based access:
- Institutional License: LKR 10,000 – 15,000 per year
- Pay as You go Plans for Individual Users
- API access for research systems: LKR 4,000 – 5,000 per month

THANK YOU
